An ordered latent field for a cloud-resolving squall line, and what a PDE for its leading channel says

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Abstract

We emulate two-dimensional CM1 squall lines with a *local* latent model: each 2 km patch of the domain becomes a site carrying 32 latent channels, and a stencil law advances the field one minute at a time. Training windows are centred on the storm half of the time, and a structured (nested) dropout orders the channels by importance, so that channel 0 alone reconstructs the full 434-variable column state to nMSE = 0.30, where an ordinary latent needs about 12 channels. Channel 0 is a cold-pool/convection coordinate. Sparse regression on it finds that its evolution is transport, not local growth: a law with spatial derivatives explains 21–42% of its held-out tendency from channel 0 alone, a local ODE explains nothing. Rewritten in physical units through a fixed map to surface θ', the leading law is an advection whose speed is linear in the cold-pool temperature deficit, $c(\theta') = 0.20 + 0.049\,\theta'\,\mathrm{km\,min^{-1}}$, with a tight and test-reproducible slope, and an anti-diffusive term. Its speed at the gust front matches the measured cold-pool speed on average, and a second, independently trained autoencoder gives the same law within about 20%. Two pre-registered tests of physical meaning fail, however: the speed does not track storm-to-storm differences, and its dependence on θ' is 3.5× steeper than density-current scaling.

1 Data and model

Data. CM1 r21.1 two-dimensional (x–z) squall lines, 400 km wide with $\Delta x = 250$ m, 62 stretched levels, open lateral boundaries and a grid moving at $12.5\,\mathrm{m\,s^{-1}}$; Morrison two-moment microphysics; 6 h runs with 1-min output. 121 storms from three ensembles (control sounding with random initial noise; and boundary-layer moisture × wind-shear variations), split by storm into training, validation and test sets. Each column holds 7 variables (u', w, θ', q_v', p', $\sqrt{q_{\mathrm{liq}}}$, $\sqrt{q_{\mathrm{ice}}}$) on 62 levels, as perturbations from the far-field initial column, standardized per variable: 434 numbers per grid column, 1 600 columns.

Channels and patches. The autoencoder is a 1D convolution along x with the column as the channel dimension (Fig. 1):

- **Patchify**: a convolution with kernel = stride = 8 maps each 8-column patch (2 km) to a hidden vector of width 256. The domain becomes $P = 200$ *sites*.
- **Mix**: two residual convolutions of radius 1 along the sites, so each site sees ±2 neighbouring sites (±4 km); no circular padding (the boundaries are open).
- **Project**: a 1×1 convolution without bias to $c = 32$ latent channels per site.

The latent state is therefore a *field* $z_j(t) \in \mathbb{R}^{32}$, $j = 1..200$, not a single vector. Shifting the input by a whole patch shifts the latent by one site. The decoder mirrors the encoder (transposed convolution, kernel = stride = 8).

Propagator: an explicit latent PDE. The model used here advances the field with a local law, one frame (1 min) per step, integrated with a Heun step:

$$\partial_t z = A_0 z + A_1 \partial_x z + A_2 \partial_x^2 z + b + Q_1[z_i z_j] + Q_2[z_i\, \partial_x z_j] + R(z, \partial_x z, \partial_x^2 z), \tag{1}$$

with centred differences over sites, a small pointwise MLP residual R, and a soft tendency cap. All coefficients start at zero (identity step). The residual ends up carrying only 1–3% of the tendency, so this 32-channel law forecasts CM1 as well as a neural stencil network. But it is dense: tens of thousands of its 52 736 coefficients are needed, and no readable law can be pruned from it. That is the motivation for Section 4: ask for a short law for the *leading* channels of an *ordered* latent instead.

Training. Stage A trains the autoencoder (reconstruction, plus a log-determinant term against channel collapse). Stage B trains the propagator on latent rollouts of up to 45 steps with the autoencoder frozen. Stage C fine-tunes both jointly on decoded 45-step rollouts, with the 3-site halo of every window forced by the true state.

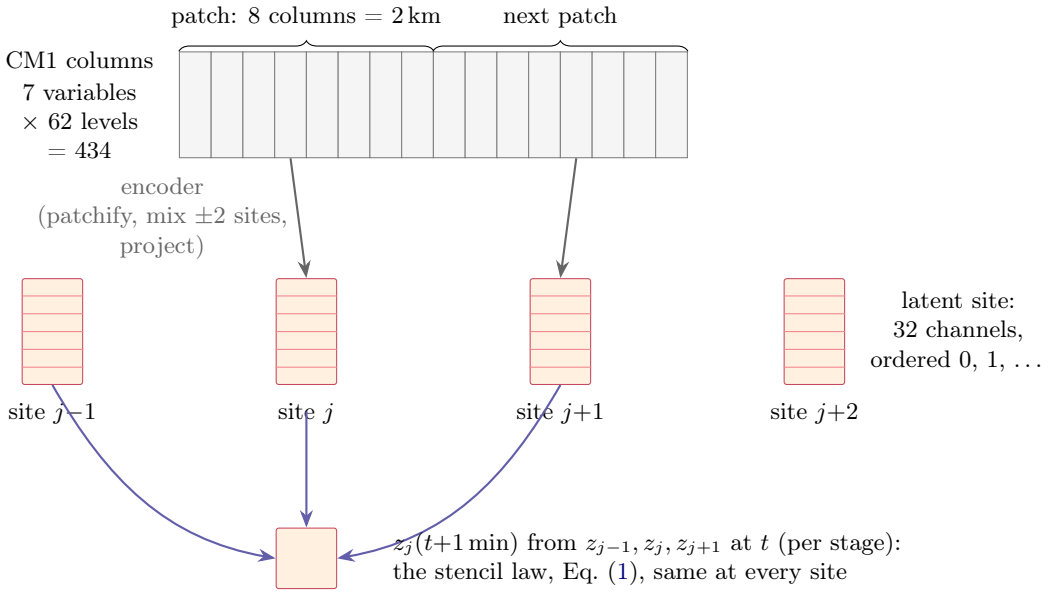


Figure 1: The local latent model. Each 2 km patch of CM1 columns is encoded into one site with 32 channels; nested dropout orders them so that channel 0 carries most of the column. A shared stencil law advances every site from its neighbours.

2 Storm-centred training

Most of the 400 km domain is quiet environment most of the time: a squall line occupies a few tens of kilometres. Uniformly sampled windows would spend most of the training signal on near-zero perturbations. Every stage therefore draws half of its training windows (`active_frac` = 0.5) centred on a storm column, a randomly chosen grid column and time with composite reflectivity above 10 dBZ; the other half is drawn uniformly, so the quiet environment and the storm's surroundings are still learned (Fig. 2). Windows are 32 sites (64 km) for the autoencoder, 48 sites (96 km) × up to 47 frames for the propagator. Because the model is local, a window is a complete training example: the same weights are applied to the whole 200-site domain at test time.

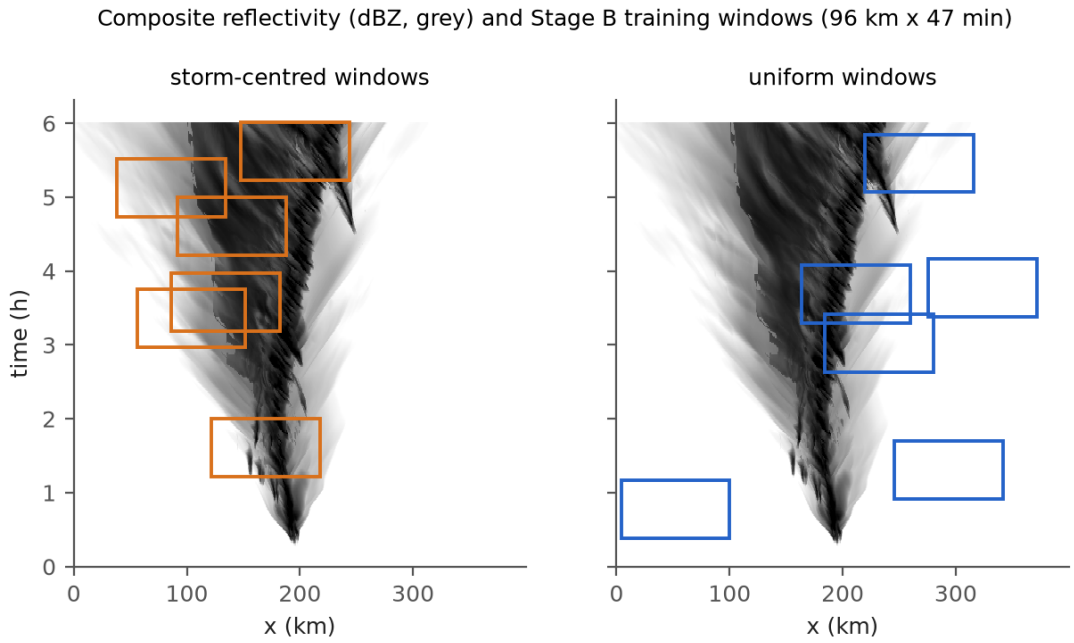


Figure 2: Composite reflectivity of one training storm (grey) with six storm-centred (left) and six uniform (right) Stage B windows. Half of all windows are storm-centred; the uniform half covers the environment.

3 Structured (nested) dropout: an ordered latent

In an ordinary autoencoder the 32 channels are interchangeable: any rotation of the latent is equally good, so no single channel means anything. Nested dropout breaks this symmetry. For half of the training samples, an integer k is drawn uniformly from 1..32, channels k..31 are set to zero, and the decoder must reconstruct the column from channels 0..$k-1$ alone. Channel 0 is then trained to be the best one-channel description of the column, channels 0–1 the best pair, and so on: a nonlinear analogue of a principal-component ordering. The truncation acts on the reconstruction terms of Stage A and Stage C; the propagator always sees all channels.

The ordering is strong (Fig. 3): channel 0 alone reconstructs the column to nMSE = 0.30, and 4 channels to 0.14. The ordinary latent needs about 12 channels to reach 0.5 with its first channels. With all 32 channels the two latents reconstruct about equally well (0.031 against 0.026), so the ordering costs little. A second nested autoencoder trained from scratch with a different seed gives the same curve to within 0.01.

What channel 0 is. On the test storms, channel 0 correlates +0.80 with surface θ', −0.86 with column condensate, −0.72 with column-maximum w and −0.46 with low-level u' (site means). Decoding a field in which only channel 0 is set shows what it encodes (Fig. 4): negative values are a mature convective column, with a surface cold pool of −8 K, latent warming aloft, liquid water below the melting level and an ice anvil above, and a reversal of u' with height (low- and mid-level flows of opposite sign, $\sim$10 m s^{-1}). Positive values are the undisturbed environment. In space and time (Fig. 6, first two panels) channel 0 tracks the cold pool and the gust front. It is a cold-pool/convection coordinate. Channels 1–3 have no comparably clear physical correlate ($|r| \leq 0.46$).

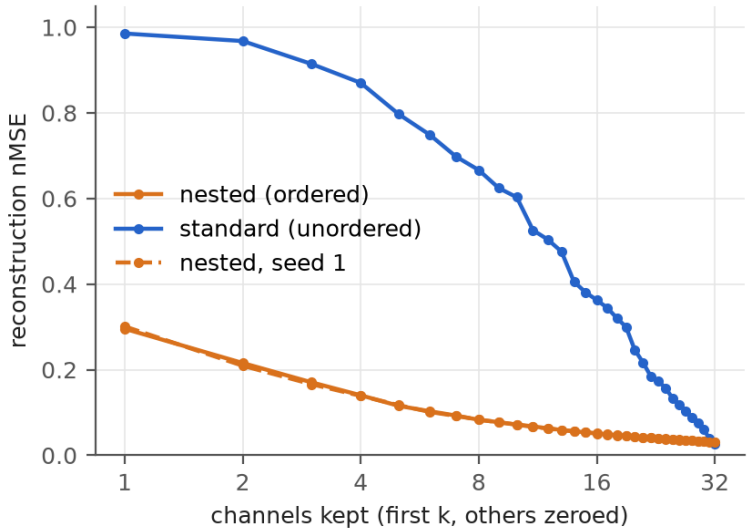


Figure 3: Reconstruction error with only the first k channels kept (others set to zero), on unseen test storms. Nested dropout orders the channels by importance; the ordinary latent spreads information evenly.

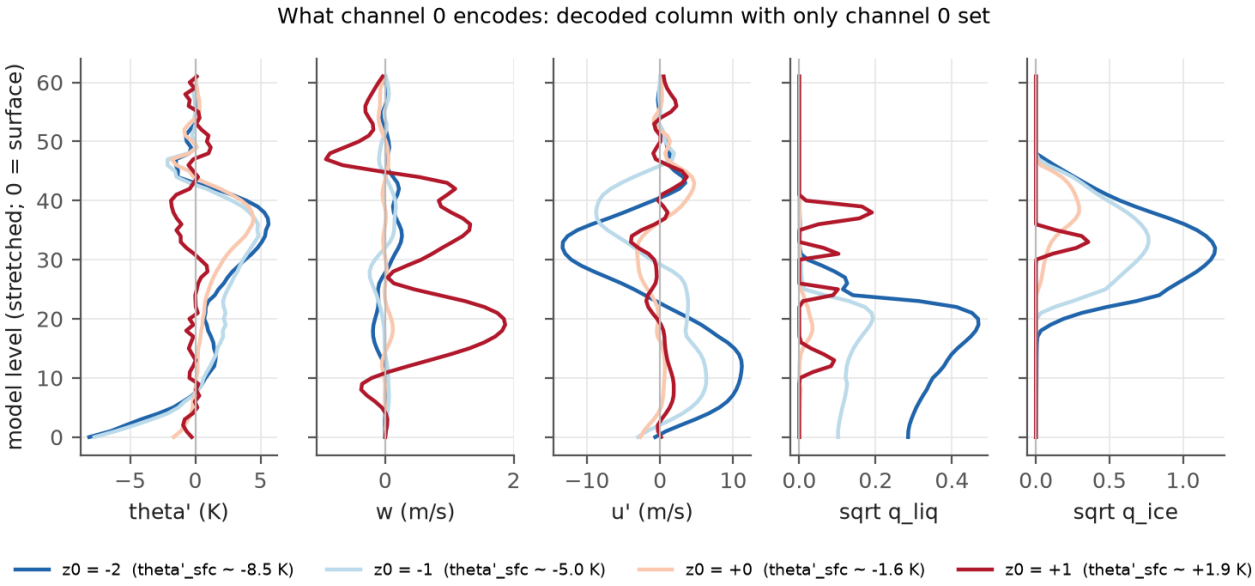


Figure 4: Decoded column at a site when only channel 0 is non-zero, for four values of z_0; the surface θ' each value maps to (Sec. 4.1) is in the legend. Condensate is plotted as $\sqrt{q}$ $(\mathrm{g\,kg^{-1}})^{1/2}$.

4 Interpretable PDEs for the leading channels

Method. For the first k channels of the ordered latent ($k = 1, 2, 4, 8$) we fit $\partial_t z_j$ (centred difference, per minute) for $j < k$ by bootstrap sequentially thresholded least squares (STLSQ, resampling whole storms) on 30 training storms, frames 30–330 min, and score on 10 test storms. The candidate libraries are built from the first k channels only:

- **local ODE**: z and quadratic products (no neighbours);
- **PDE**: plus $\partial_x z$, $\partial_x^2 z$ (and optionally up to $\partial_x^4 z$) and their products with z;
- options: products of two derivatives (KPZ-type), cubic terms, second order in time, and a BBM-type mixed operator $(1 - \ell^2 \partial_x^2)\partial_t z$.

Coefficients are in sites (2 km) and minutes. Decision rules for each test were written down before the corresponding run.

Channel 0 is transported, not locally driven. Fig. 5 shows the held-out R^2 of channel 0's tendency. A local ODE explains essentially nothing at any k ($R^2 \leq 0.2$); spatial derivatives explain 21% from channel 0 alone (42% with third and fourth derivatives), rising to 56–61% with two channels and 76–82% with eight. On the ordinary (unordered) latent the same procedure finds almost nothing ($R^2 \leq 0.27$): the ordering is what makes a leading-channel law possible.

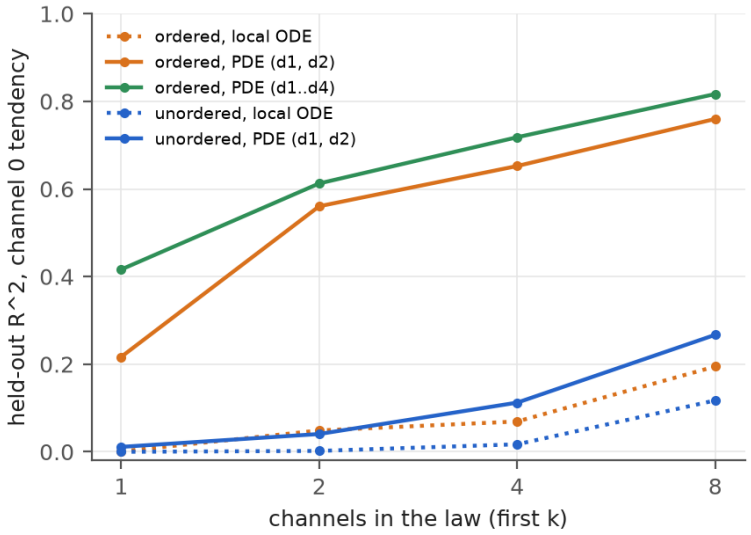


Figure 5: Held-out R^2 of channel 0's tendency against the number of channels in the law (threshold 0.03). Orange/green: ordered latent; blue: ordinary latent. Dotted: local ODE; solid: with spatial derivatives.

The shortest laws. With channel 0 alone, the sparsest PDE (3 terms) is a nonlinear advection,

$$\partial_t z_0 = -(0.059 + 0.082\, z_0)\, \partial_x z_0 - 0.004. \tag{2}$$

With more channels, channel 0's equation keeps the same form, but its advecting velocity is built from several channels: for $k = 8$ its 13 terms are almost all of the form $z_i\, \partial_x z_j$, led by $\partial_t z_0 \approx (0.069\, z_1 - 0.064\, z_0)\, \partial_x z_0 - 0.050\, z_1 \partial_x z_1 + \ldots$ (Table 1). The selected structure is stable across bootstrap resamples. Other library extensions add little: a $(\partial_x z_0)^2$ term (KPZ-type) is selected with a consistently negative sign; cubic terms and second order in time do not help; third-derivative terms raise R^2 but their coefficient doubles when the field is smoothed by one more site, so they describe the unresolved sharpness of the gust front rather than a physical dispersion.

Table 1: Selected terms of channel 0's equation (one-step fit, threshold 0.1; coefficients in sites and minutes; bootstrap selection frequency 1.0 unless noted).

law	terms	held-out R^2
$k = 1$, PDE	$-0.082\, z_0 \partial_x z_0$, $-0.059\, \partial_x z_0$ (0.9), -0.004	0.21
$k = 1$, PDE + $(\partial_x z)^2$	$-0.095\, z_0 \partial_x z_0$, $-0.074\, \partial_x z_0$ (0.7), $-0.046\, (\partial_x z_0)^2$ (0.7), … (7 terms)	0.25
$k = 8$, PDE	$+0.069\, z_1 \partial_x z_0$, $-0.064\, z_0 \partial_x z_0$, $-0.051\, z_1 \partial_x z_1$, $+0.026\, z_1 \partial_x z_5$, … (13 terms)	0.65

4.1 The leading law in physical units

A coefficient on z_0 is not physical by itself: the sign, scale and zero point of a latent channel are arbitrary (a gauge). For example, the constant speed in Eq. (2), $0.12\,\mathrm{km\,min^{-1}}$, is the speed at $z_0 = 0$ and changes if z_0 is shifted. We fix the gauge with a physical quantity: on the training storms, $\theta'_{\mathrm{sfc}} = s\, z_0 + b$ with $s = 3.48\,\mathrm{K}$ and $b = -1.55\,\mathrm{K}$ ($r = 0.79$; the OLS and inverse-OLS slopes, 2.76 and 4.38 K, bound the uncertainty of s). Fitting the 4-term law $\partial_t z_0 = k_0 + a_1 \partial_x z_0 + a_2 z_0 \partial_x z_0 + \nu \partial_x^2 z_0$ by ordinary least squares (no sparsity, no tuning for stability) and substituting gives

$$\partial_t \theta' = -c(\theta')\, \partial_x \theta' + \nu\, \partial_x^2 \theta' + S, \qquad c(\theta') = c_0 + c_1 \theta', \tag{3}$$

	R^2	c_0 ($\mathrm{km\,min^{-1}}$)	c_1 ($\mathrm{km\,min^{-1}\,K^{-1}}$)	ν ($\mathrm{m^2\,s^{-1}}$)	c at the front ($\mathrm{km\,min^{-1}}$)
training storms (30)	0.19	+0.202	+0.0490	−614	-0.277 at $\theta' = -9.8\,\mathrm{K}$
bootstrap 95% CI		[0.168, 0.236]	[0.0462, 0.0517]	[−671, −551]	[−0.313, −0.232]
test storms (10)	0.21	+0.173	+0.0476	−628	-0.303 at $\theta' = -10.0\,\mathrm{K}$
seed-1 autoencoder, training	0.24	+0.225	+0.0578	−963	-0.336 at $\theta' = -9.7\,\mathrm{K}$
bootstrap 95% CI		[0.179, 0.273]	[0.0548, 0.0610]	[−1022, −898]	[−0.380, −0.286]
seed-1 autoencoder, test	0.27	+0.190	+0.0567	−940	-0.373 at $\theta' = -9.9\,\mathrm{K}$

The law says (Fig. 7a):

- **The colder the air, the faster it moves in the storm's direction.** The speed is linear in θ' and crosses zero at $\theta' \approx -4\,\mathrm{K}$; at the gust front ($\theta' \approx -10\,\mathrm{K}$) it is -0.28 to $-0.30\,\mathrm{km\,min^{-1}}$, equal to the measured mean speed of the cold-pool centroid (-0.29). The slope c_1 is determined to ±6% and the test storms reproduce it to 3%. A local estimate (the advection speed fitted separately in 2 K bins of θ', grey) is close to the line from -16 to $-6\,\mathrm{K}$ and flattens above.
- **The diffusion is negative** ($\nu \approx -600\,\mathrm{m^2\,s^{-1}}$, CI excluding zero): channel 0's one-step law sharpens the front. As a PDE it is ill-posed, which explains why the closed rollouts of most unrefitted laws blew up, and why refitting the laws on rollouts (not shown) adds large positive diffusion to stay stable. That diffusion is a stability device, not eddy mixing (the physical-field pilot gives $\sim 30\,\mathrm{m^2\,s^{-1}}$).
- **It is one fifth of the story**: $R^2 = 0.2$. The rest of channel 0's tendency needs the other channels (Fig. 5). Fig. 6 compares the law's tendency with the true one: the law captures the advance of the gust front, not the convective cells behind it.

Does it depend on the autoencoder? A second nested autoencoder, trained from scratch with another seed, is a different latent coordinate system: its channel 0 even has the opposite sign ($\theta'_{\mathrm{sfc}} = -3.46\, z_0 - 1.61\,\mathrm{K}$, $r = -0.81$). After the same gauge fixing, its law has the same form and signs (table, dashed line in Fig. 7a): c_1 is 18% larger and ν 57% more negative, and the two bootstrap intervals for c_1 do not overlap. The law's structure is a property of the data mapped through channel 0; its precise coefficients carry an autoencoder-dependent spread of roughly 20%.

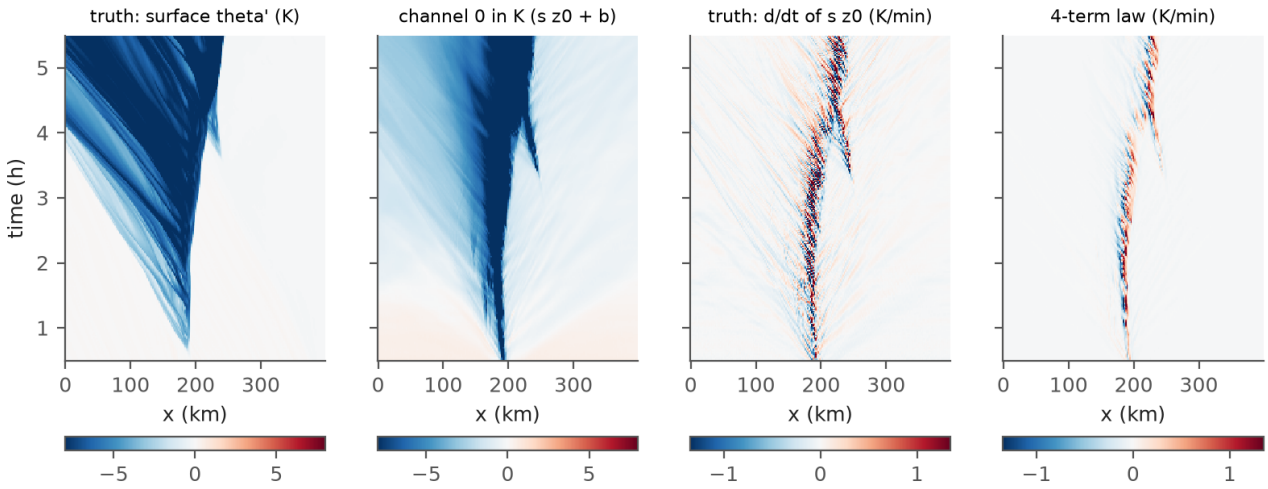


Figure 6: One test storm in x–t. From left: true surface θ'; channel 0 mapped to kelvin; the true tendency of the mapped channel 0; the 4-term law's tendency.

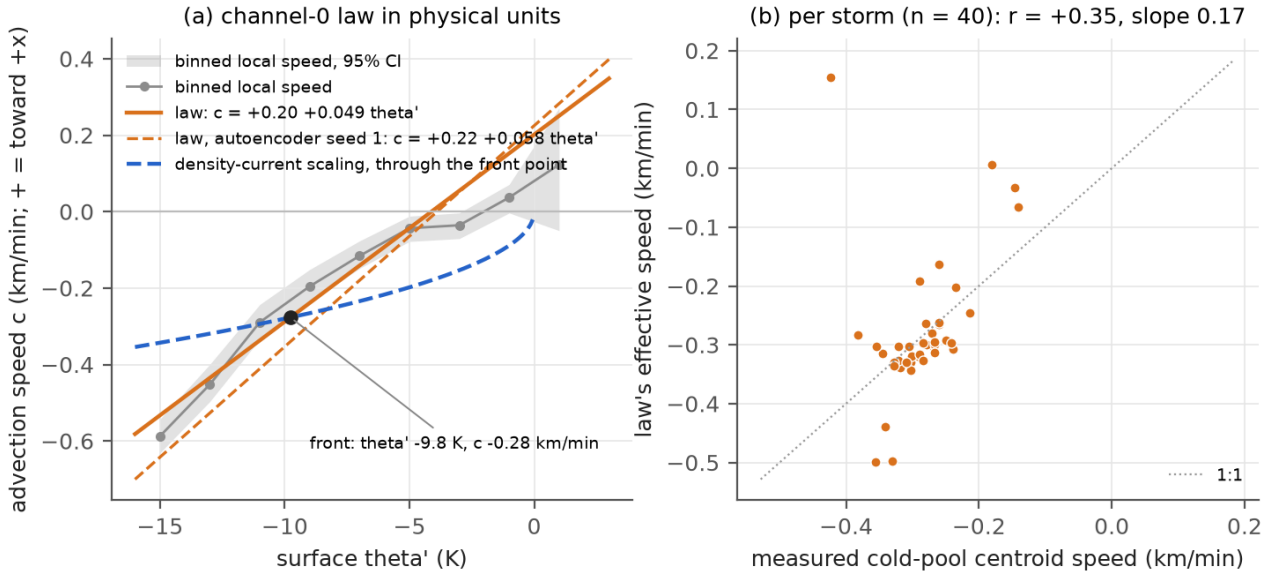


Figure 7: (a) Advection speed of the channel-0 law against surface θ' (orange), the speed fitted locally in θ' bins (grey, bootstrap over storms), and density-current scaling $|c| \propto \sqrt{|\theta'|}$ drawn through the front point (blue, dashed). (b) Per storm: the law's effective speed, fitted on each storm alone, against the measured cold-pool centroid speed.

Pre-registered tests of physical meaning. Two tests were fixed before the runs.

1. **Does the law's speed track each storm's speed?** Fitting the advection law on each of 40 storms and comparing with the storm's measured speed: Pearson $r = 0.35$ (cold-pool centroid) and 0.40 (condensate centroid), 95% CIs covering zero, and a slope of 0.2 against the required 0.5–2 (Fig. 7b). **Not met**: the law gets the mean speed right but not the storm-to-storm variation. (With the field smoothed by one site: $r = 0.46$–0.49, slope 0.35; with the seed-1 autoencoder: $r = 0.31$, slope 0.12. Not met.)
2. **Is the θ' dependence density-current-like?** The rule required $c_0 < 0$, $c_1 > 0$ with a CI excluding zero, c_1 within a factor 3 of the density-current prediction $d|c|/d|\theta'| = |c|/(2|\theta'|) = 0.014\,\mathrm{km\,min^{-1}\,K^{-1}}$, the test value inside the training CI, and agreement for an independently trained autoencoder. The sign of c_1, the test agreement and the seed-1 agreement (same signs, c_1 within a factor 2) are met; but c_0 is positive (ambient air moves the other way), and c_1 is 3.5× the density-current value (2.7–4.4× over the bounds on s; 3.3× for seed 1). **Not met.**

5 Summary

- A local latent field (32 channels per 2 km site) with storm-centred training emulates CM1 squall lines; its own dynamics is an explicit, but dense, 32-channel PDE.
- Nested dropout gives an importance-ordered latent at no cost in accuracy. Its channel 0 alone carries 70% of the column variance and is a cold-pool/convection coordinate; without the ordering no channel has a physical meaning.
- The leading channel's evolution is transport. In physical units its shortest law is an advection with a speed linear in the cold-pool deficit, a reproducible slope, the right mean front speed, and anti-diffusion.
- The coefficients do not pass the stricter physical tests: they do not track storm-to-storm speed differences, and the θ' dependence is not density-current scaling. The law is an effective, gauge-fixed description of how the latent front moves, not a recovered physical equation.